

Openjudge: An Online Judge for Offline Programming Contests



Do you need an impartial judge for a programming contest? Openjudge is a LAN based program judging implementation in Python. Get a taste of it in this introductory article.

Almost all institutes have a computer science department or an IT department, nowadays. Very often, these departments organise programming contests. While participating in these contests, I became aware of the following things:

- Eighty per cent of the time, the programs are checked manually
- A lot of people participate
- People prefer the contests to be online (like codechef.com)
- Offline events have their own charm and draw large crowds

The advantages offered by online program judges are numerous:

- Multiple language support
- Quick checking
- No biases

Openjudge bridges the gap between online judges and offline contests. It offers all the capabilities of an online

judge for an offline system, with very few requirements. One can set up a programming competition with minimal effort spent in developing the infrastructure needed for it. So let us get started.

The set-up

Let us explore the individual components of Openjudge. To install it, we first need some configuration. Openjudge was written with Linux in mind in order to eliminate complex issues arising from supporting multiple operating systems. This article was written using Linux Mint 17. It is known to work on Ubuntu also. The prerequisites are:

1. Linux Mint/Ubuntu
2. Python 3
3. Git
4. Pip and Virtualenv

In order to complete this list of requirements, let's first get